

## Double Columnar Transposition Cipher

### Code:

```
import numpy as np

def columnar_transpose_encrypt(text, key):
    key_order = sorted(range(len(key)), key=lambda k: key[k])
    num_rows = -(-len(text) // len(key))
    grid = np.full((num_rows, len(key)), 'X', dtype=str)
    for i, char in enumerate(text):
        grid[i // len(key), i % len(key)] = char
    ciphertext = "".join("".join(grid[:, i]) for i in key_order)
    return ciphertext

def columnar_transpose_decrypt(ciphertext, key):
    key_order = sorted(range(len(key)), key=lambda k: key[k])
    num_rows = -(-len(ciphertext) // len(key))
    grid = np.full((num_rows, len(key)), 'X', dtype=str)
    index = 0
    for i in key_order:
        for j in range(num_rows):
            if index < len(ciphertext):
                grid[j, i] = ciphertext[index]
                index += 1
    return "".join("".join(row) for row in grid).rstrip('X')

text = input("Enter the plain Text: ")
key = input("Enter the key: ")
cipher = columnar_transpose_encrypt(text, key)
print("Encrypted:", cipher)
print("Decrypted:", columnar_transpose_decrypt(cipher, key))
```

```
double_columnar.py X
double_columnar.py > ...
14 def columnar_transpose_decrypt(ciphertext, key):
20     for i in key_order:
21         for j in range(num_rows):
22             if index < len(ciphertext):
23                 grid[j, i] = ciphertext[index]
24                 index += 1
25
26     return "".join("".join(row) for row in grid).rstrip('X')
27
28 text = input("Enter the plain Text: ")
29 key = input("Enter the key: ")
30 cipher = columnar_transpose_encrypt(text, key)
31 print("Encrypted:", cipher)
32 print("Decrypted:", columnar_transpose_decrypt(cipher, key))
33
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 6

PS C:\Users\thava\OneDrive\Desktop\Currently working\SC> & C:/Users/thava/AppData/Local/Programs/Python/Python39-64/Scripts/python C:\Users\thava\OneDrive\Desktop\Currently working\SC\double\_columnar.py  
Enter the plain Text: Thavamani  
Enter the key: THiskey  
Encrypted: hiTnmXaXaXvXaX  
Decrypted: Thavamani